HybridEP + Flex Dispatcher

One MoE dispatch API, a few-SM backend for any NVLink domain

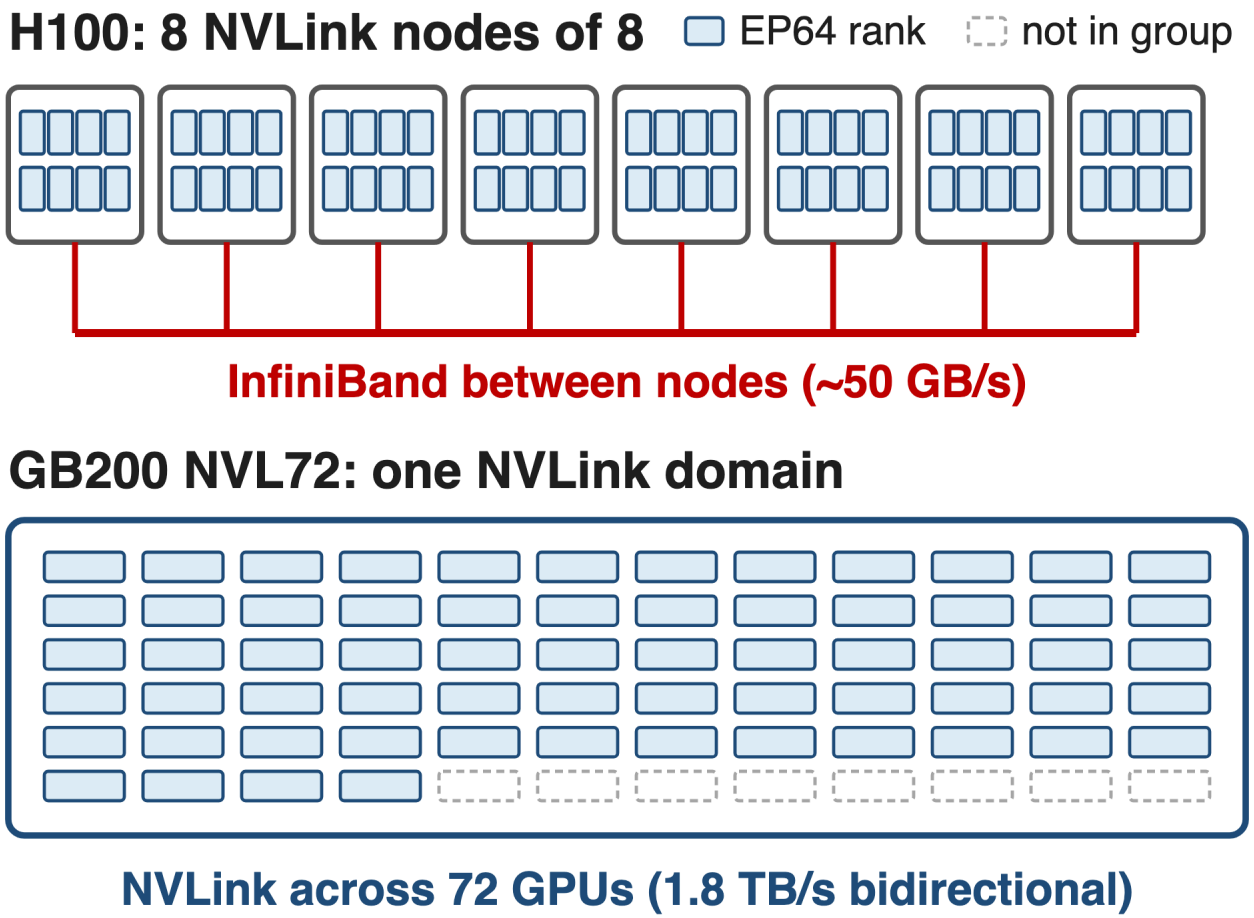
NVIDIA · Megatron-Core MoE (v0.15+) · DeepEP hybrid-ep branch

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Expert-Parallel Communication · arxiv.org/abs/2603.07685

EP64 spans eight NVLink islands or one domain

- DeepSeek-V3: **256** experts, top-8
- **116** dispatch/combine ops per forward pass
- One MoE code path must serve both networks



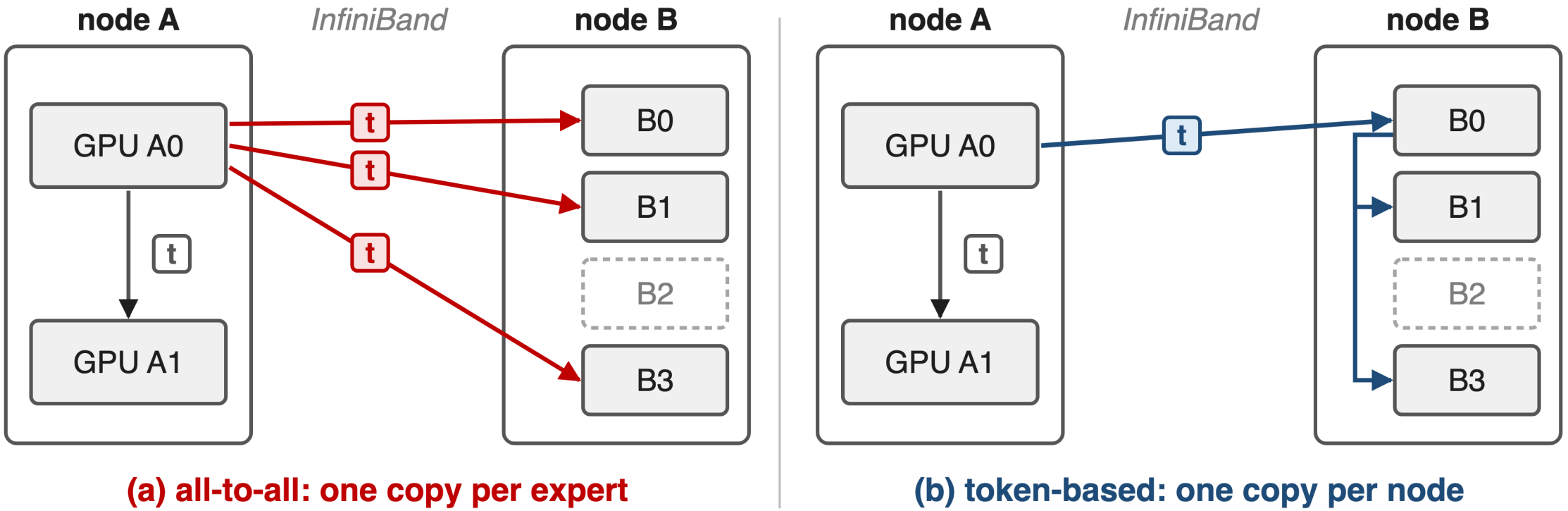
Token exchange eats up to 60% of training

40–60%

of DeepSeek-V3 training time is EP all-to-all when EP64 crosses H100 nodes (≈20% even inside GB200 NVL72)

Before optimization. Megatron-Core MoE report, arXiv 2603.07685, §4.2 (p.29)

All-to-all ships one copy per expert, not per node



Plus a host sync for split sizes and separate permute kernels

DeepEP drops duplicates but costs SMs

- **NCCL all-to-all**
- **top-k copies** on the wire
- Host sync, separate permute kernels

- **DeepEP (v1)**
- **20 SMs**/GPU: ~**20%** GEMM efficiency lost
- **≤ 8 NVLink peers**: no NVL72 domain

Needed: one copy per domain, few SMs, any NVLink size, one API

Insight: engines move bytes, SMs only orchestrate

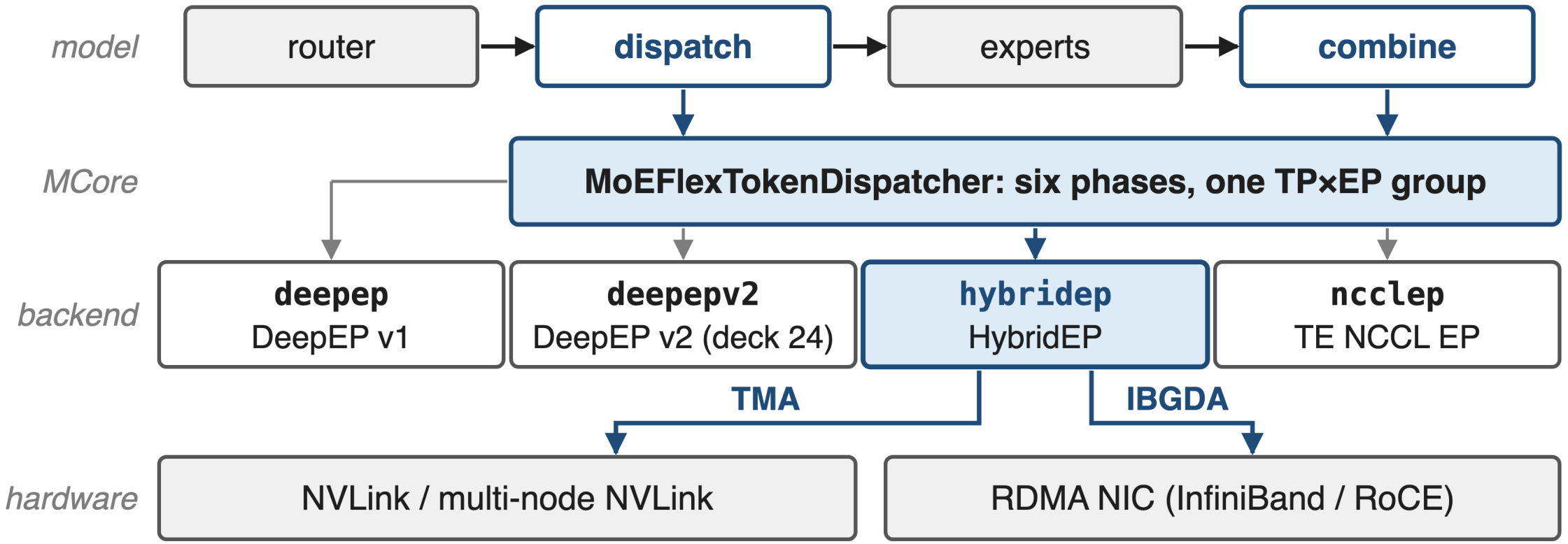
- **TMA**: bulk copies, shared memory ↔ local or peer HBM
- **IBGDA**: a GPU warp posts RDMA work to the NIC itself

Insight: engines move bytes, SMs only orchestrate

- **TMA**: bulk copies, shared memory ↔ local or peer HBM
- **IBGDA**: a GPU warp posts RDMA work to the NIC itself
- Measured: **8** SMs fill NVLink (DGX H100); **~4** approach a 400 Gb/s NIC

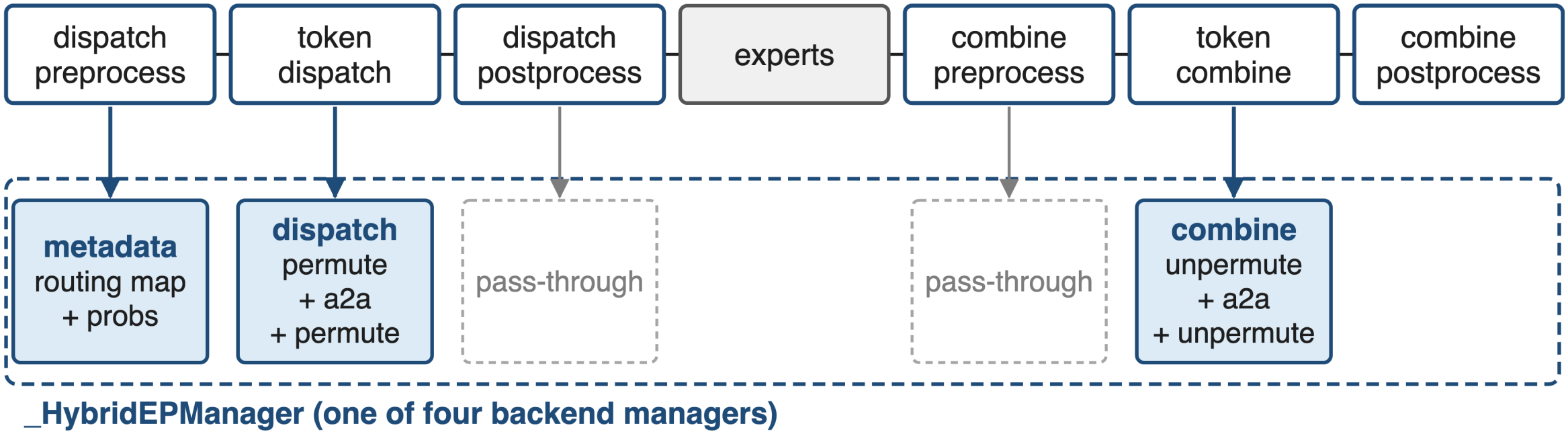
A few SMs suffice when each runs a whole copy pipeline

One Flex dispatcher fronts four backends



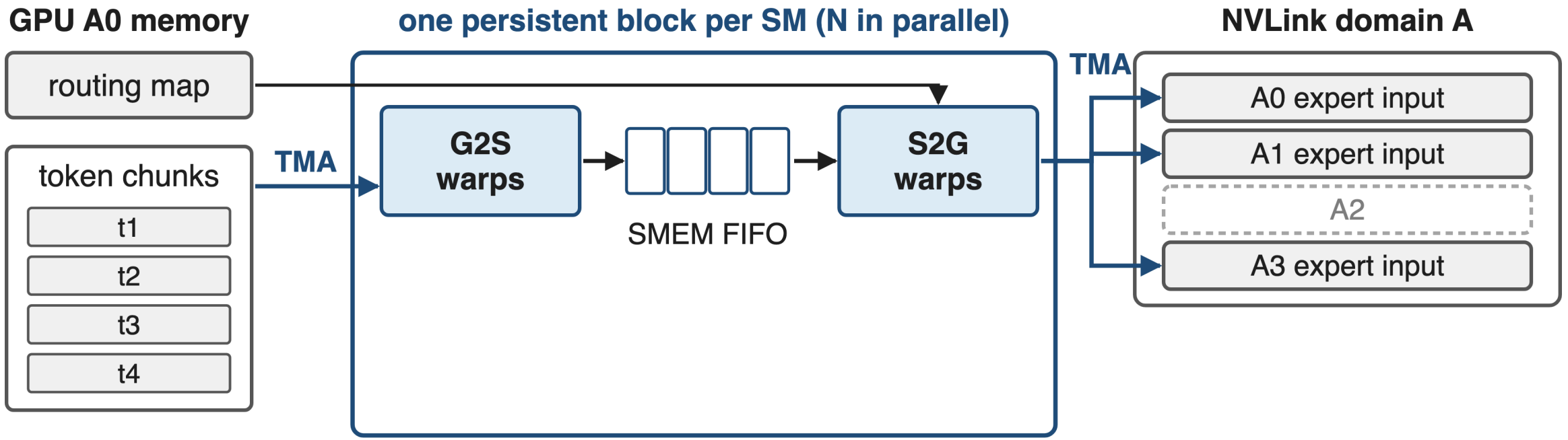
--moe-token-dispatcher-type flex --moe-flex-dispatcher-backend deepep | deepepv2 | hybridep | ncclep (HybridEP added in MCore v0.15)

❶ Six phases stay fixed; only the manager changes



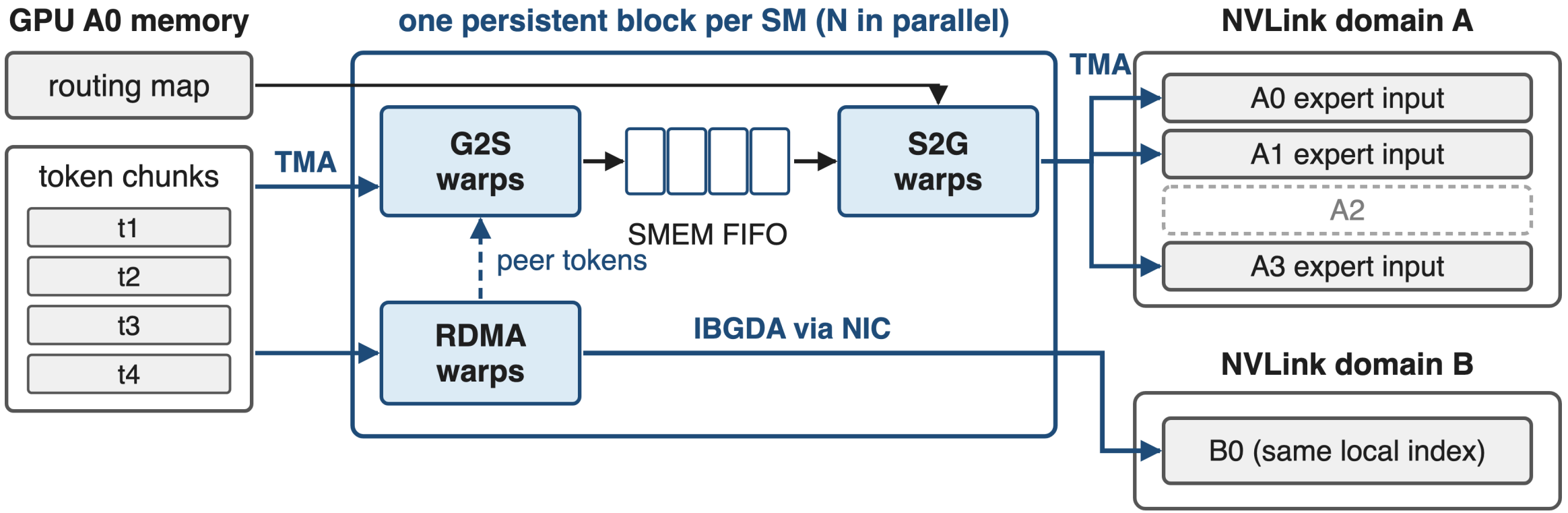
The routing map spans one TP×EP group: backends never see TP or EP

❷ Dispatch: a TMA pipeline in every SM



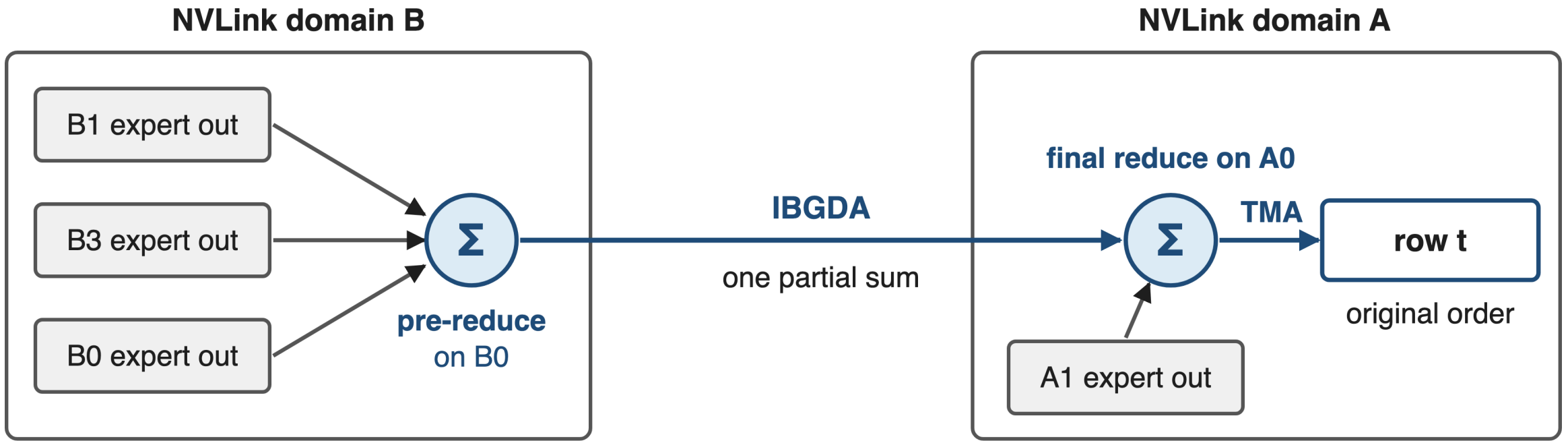
One NVLink domain (for example NVL72): G2S loads chunks, S2G stores each token only to ranks the routing map selects

❷ Dispatch: a TMA pipeline in every SM



Several domains (H100): one RDMA copy per remote domain, sent to the same-index GPU, then forwarded over NVLink

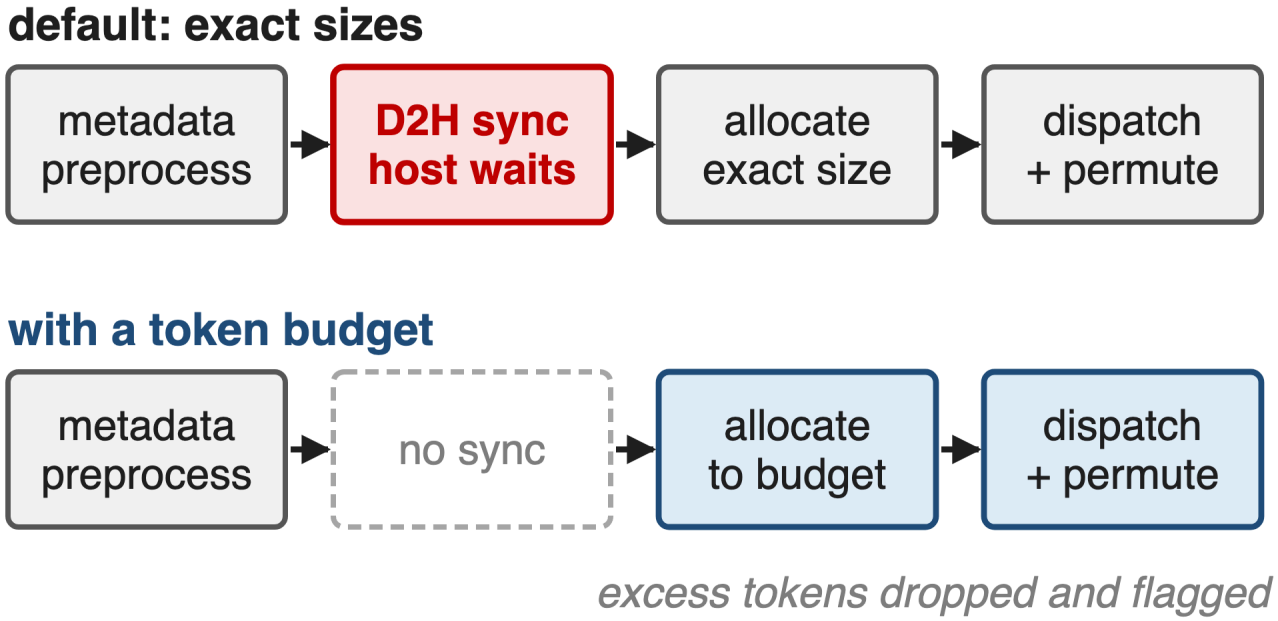
❸ Combine reduces inside the kernel, twice



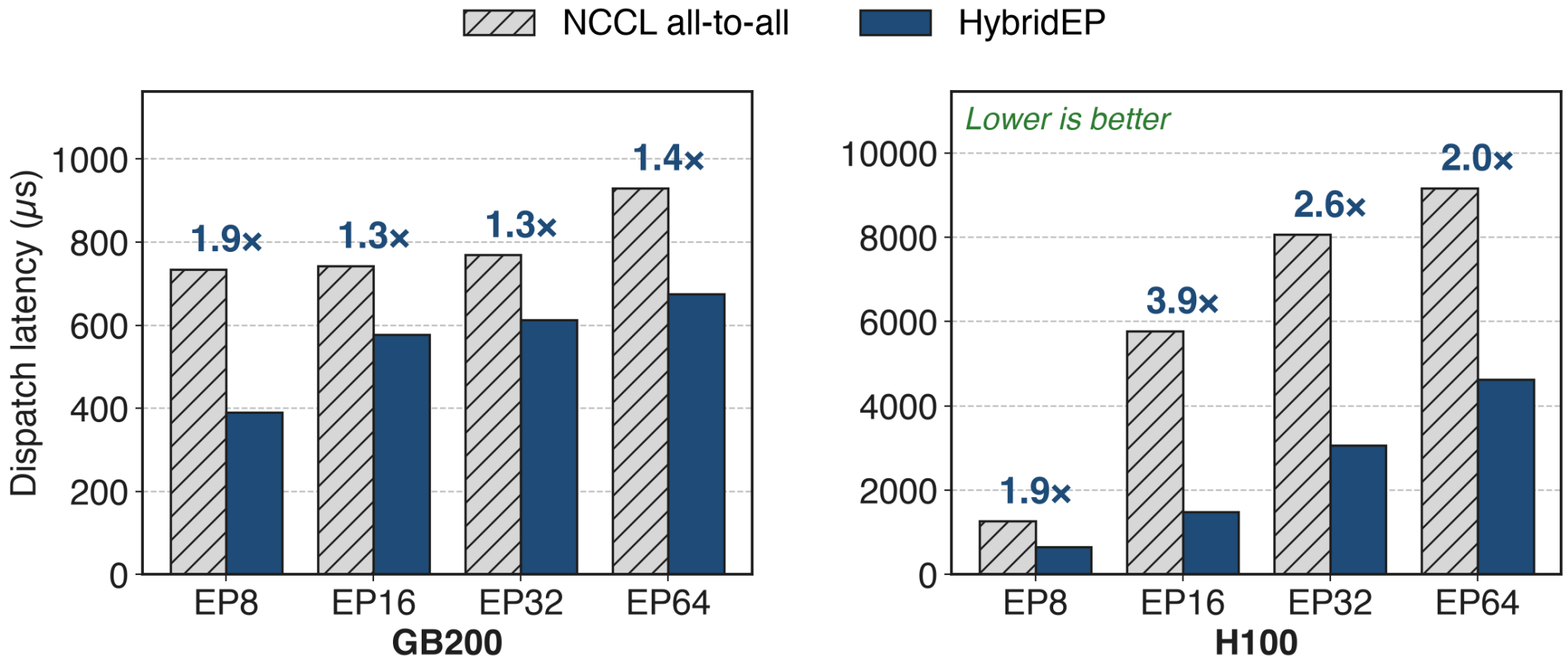
One partial sum crosses the network per domain; no separate unpermute pass

❹ Buffers sized up front remove the host sync

- Init finds NVLink-reachable ranks: NVL72 EP64 is **one** domain
- One worst-case registered buffer per rank
- Token budget: no host sync, graph-capturable (deck 38)

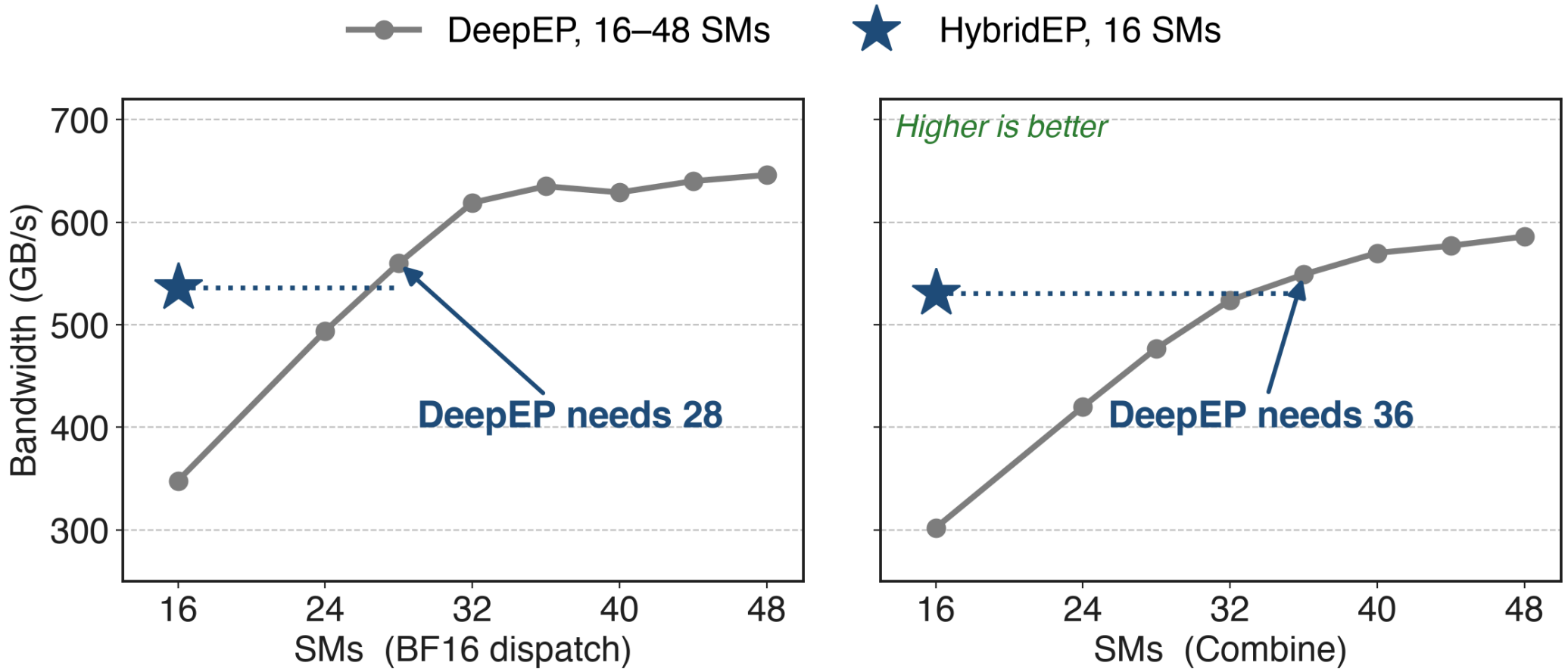


HybridEP dispatch is 1.3–3.9× faster than all-to-all



arXiv 2603.07685 Table 7, communication latency only; hidden 7168, sequence 4096, 256 experts. Ratios derived. Combine: 1.1–3.3×.

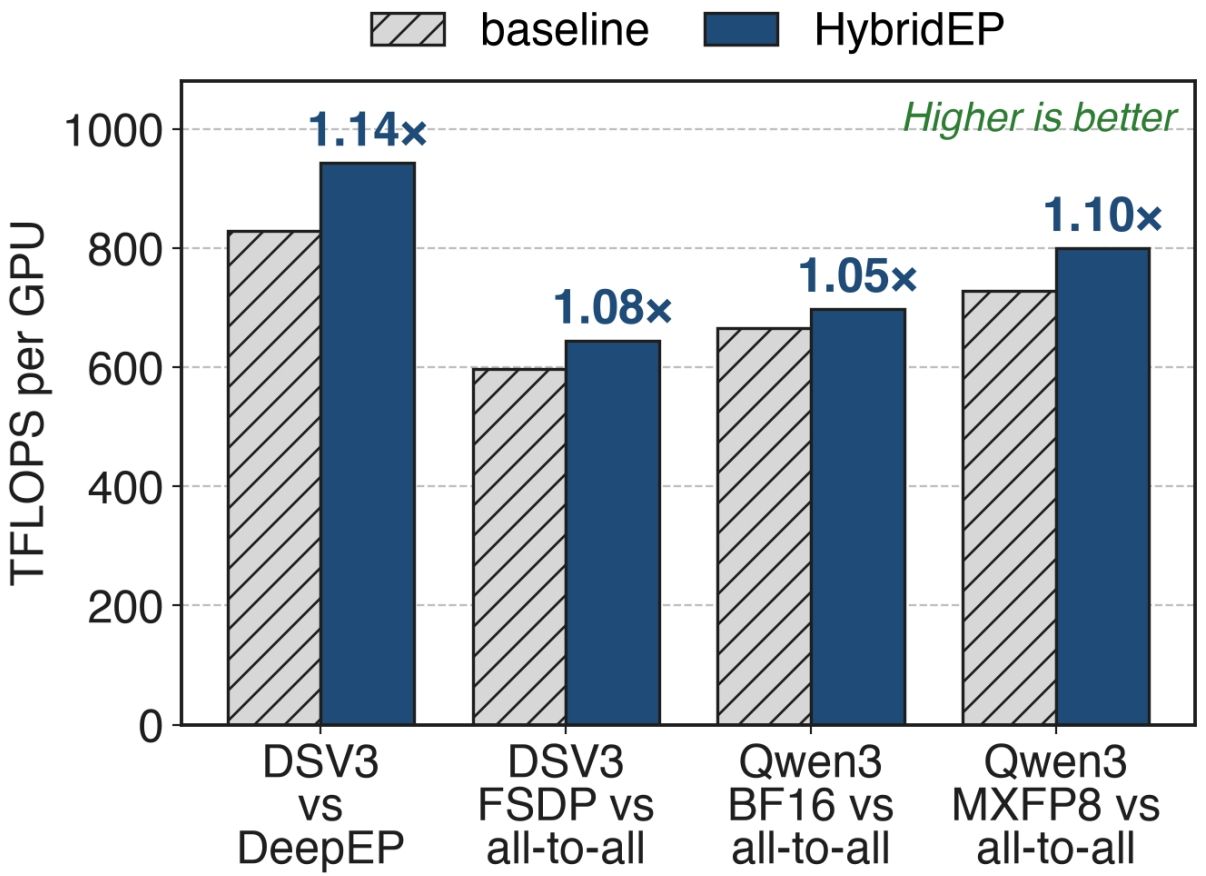
16 HybridEP SMs match 28–36 DeepEP SMs



B200, EP8 in one NVLink domain, Torch-API bandwidth; 4096 tokens, hidden 7168, top-8 (Hybrid-EP README). SM thresholds derived.

HybridEP lifts DeepSeek-V3 1.14× over DeepEP

- Megatron Core on Grace Blackwell
- DeepSeek-V3: **829 → 943** TFLOPS/GPU
- vs all-to-all: **1.05–1.10×**



NVIDIA Hybrid-EP blog (Feb 2026), Table 1; DSV3 = DeepSeek-V3

HybridEP pays in memory and maturity

- **~3.75–4 GB** registered buffer per rank (EP64)
- Dispatch and combine share it: they cannot overlap
- nvcc JIT compile on first use
- Opt-in, "[Experimental]"; default backend deepep
- MCore path: BF16 dispatch, FP32 probs
- H100 best config: DeepEP + overlap (deck 11)

Summary

- EP all-to-all: **40–60%** of DeepSeek-V3 time across H100 nodes
- **Insight**: TMA and IBGDA move bytes; a few SMs orchestrate
- One Flex API; one copy per domain, fused reduce, MNNVL
- **1.3–3.9×** faster dispatch; **16** SMs ≈ **28–36**; **1.14×** end to end

https://arxiv.org/abs/2603.07685